

Join us to work on one of the most promising single-photon detection technologies! The UBC team at the [MiNa labs](#), [Optical Computing Group](#), and [Photonics & Nanostructures Laboratory](#), is on the frontier of developing novel electro-optic devices and systems that can revolutionise the computing, biosensing, and communications paradigm. Quantum optical systems encode information using single-photon states of light; thus, they require the ability to reliably generate, manipulate, and detect single photons.

Job description

This project focuses on the development of various optical components needed to perform cryogenic quantum optical signal processing. You will be working directly with cryogenic systems, assisting in the operation, opening and closing of cryostats, system configuration, and maintaining stable operating conditions in collaboration with graduate researchers. The role involves hands-on mechanical work and is best suited for students comfortable using basic tools and working with physical hardware.

You will also support the design, integration, and testing of Printed Circuit Board (PCB) electronics into a complete superconducting single-photon detection (SNSPD) system. These detectors are based on superconducting nanowires held in a cryostat at a temperature around 2 K, and since their invention in 2001, they have demonstrated great performance in terms of high detection efficiency, fast timing, and low dark count rate.

Building on an existing multi-channel PCB, the student will focus on debugging, performance characterization, and iterative design improvements, with an emphasis on reducing electronic noise and improving signal quality, ensuring compatibility with low-noise amplifiers, bias tees, and detector interfaces. The role combines practical hardware engineering with hands-on experimental work in cryogenic environments. The student will work closely with a Master's student, providing both engineering support and experimental assistance. Remuneration commensurate with qualifications.

Benefits

The successful candidate will have the opportunity to learn more about the cryogenic industry, photonics, electronic design, and quantum photo-detectors. The intern will be working with a research team that is actively involved in industrial collaborations with external companies. You will be tackling new and unsolved problems with the guidance of other senior researchers, and introduce yourself to cutting-edge R&D projects in a rapidly growing field, and the opportunity to pioneer solutions in the quantum signal processing industry.

Expected qualifications

- Familiarity & experience with basic mechanical tools & mechanical hardware;

- Experience with basic electrical testing equipment;
- Good communication, presentation, and analytical skills;
- Self-motivation, initiative, readiness to own the project, perform own research, to tackle new and unique problems;
- Ability to create concise and clear documentation and reports;
- Interest in electronics & cryogenic systems.

Desirable qualifications

- Programming experience (e.g. Python, micro-controllers, or similar tools);
- Working experience with Linux operating systems and deployment tools.